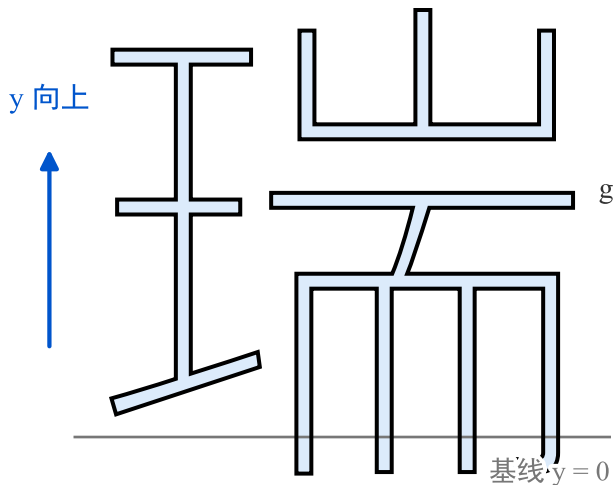


① 字体坐标系 (y 向上)



② pen.getCommands(): SVG 路径指令

glyph.draw(pen)

```
d="M912 578H418V789H447V607H643V829H672V6..."
```

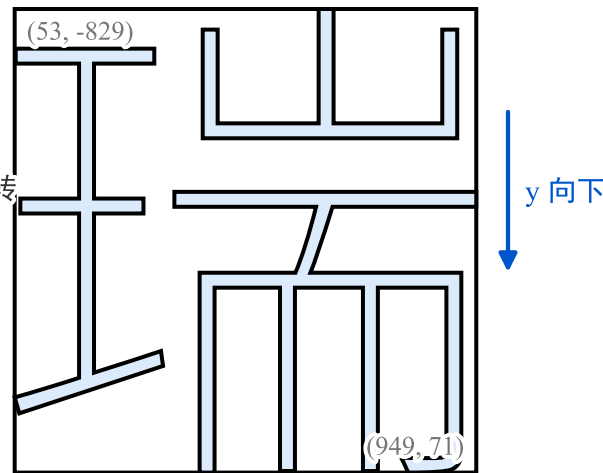
拆分指令 (共 63 条):

```
0  M912 578
1  H418
2  V789
3  H447
4  V607
5  H643
6  V829
7  H672
8  V607
9  H883
10 V789
```

..... 共 63 条, 拼成 368 字符的 d 字符串

③ SVG 视口渲染 (y 向下)

scale(1,-1) 翻转



```
<svg viewBox="53 -829 896 900">
<path d="..." transform="scale(1,-1)"/>
</svg>
```

M = moveTo (新轮廓)

L = lineTo (直线)

H V = 水平/垂直线 (lineTo 优化)

Q = qCurveTo (二次曲线)

Z = closePath (闭合)

SVG 视口 y 轴向下: 路径写入文档时需 scale(1,-1) 翻转, viewBox 起点取 (xmin, -ymax)